

Problem Statement & Exploratory Findings

Online Retail Dataset | Customer Churn & Purchase Behaviour Prediction | Mohammed Mahmoud & Mahmoud Essa

1. Real-World Problem

A UK-based online gift retailer sells exclusively via non-store channels. Despite holding a growing transaction history, the business currently applies identical post-purchase treatment to all customers, leaving significant retention opportunities unexploited. The core problem is: **the business cannot identify which customers are drifting toward churn** until it is too late to intervene cost-effectively.

The proposed IDSS addresses this by predicting, at the customer level, the probability of churn—defined as no purchase in the 90 days preceding the observation date—using transactional RFM signals and engineered behavioural features.

2. Stakeholders & Decision-Making Needs

Stakeholder	Primary Decision	IDSS Output Needed
Marketing Manager	Which customers to target with retention campaigns?	Churn probability score + risk tier
Customer Success	Who needs proactive outreach this week?	Daily ranked list of at-risk customers
CFO / Finance	What is our projected customer lifetime value?	Segment-level projected CLV and risk-adjusted customer value

3. Business Value

Retaining an existing customer costs 5–7× less than acquiring a new one. The dataset spans 12 months (December 2010 – December 2011) with **4,338 unique customers** across **37 countries**.

- Even a **10% improvement** in retention can yield an estimated **£50,000–£200,000 in additional annual revenue** (estimated from average order value and purchase frequency).
- Targeted campaigns directed by churn scores reduce wasted outreach by an estimated **30–40%**, improving marketing ROI.
- Identification of high-value at-risk customers enables differential treatment (personalised discounts, priority service), protecting the highest-margin segment.

4. IDSS Objective & Task Definition

Task Type: Binary Classification

Each customer is labelled as **Churned** = 1 (no purchase in the final 90 days) or **Active** = 0. The model outputs a probability score in $[0, 1]$ and a binary label.

Metric	Justification
F1-Score (<i>primary</i>)	Balances false negatives (missed churners → lost revenue) and false positives (wasted marketing spend). Best suited for imbalanced binary business problems.
ROC-AUC (<i>secondary</i>)	Threshold-agnostic measure of discrimination—allows the business to set its own risk tolerance post-deployment.
Recall (<i>tertiary</i>)	Slightly preferred: a false negative costs $\approx$ £300 in lost revenue vs. $\approx$ £2–5 per false positive in campaign spend.

5. Dataset Overview

The **Online Retail dataset** (UCI ID=352) contains 541,909 transactions from a UK-based non-store retailer (1 December 2010 – 9 December 2011), exposed as 6 feature columns plus **InvoiceNo** and **StockCode** as index columns. After removing returns, cancellations, and anonymous transactions (no CustomerID), **392,669 valid rows** across **4,338 unique customers** remain for modelling.

Feature	Type	Notes / Quality Issues
InvoiceNo	Categorical	Index column. Prefix C = cancellation → excluded
StockCode	Categorical	Index column. Product ID; some codes are service charges
Description	Text	$\approx$ 0.3% missing → rows dropped
Quantity	Numeric	10,600 rows negative (returns) → excluded
InvoiceDate	Datetime	Stored as string → converted to datetime
UnitPrice	Numeric	Rows with $\text{UnitPrice} \leq 0$ → excluded
CustomerID	Numeric	$\approx$ 25% missing (guest purchases) → excluded
Country	Categorical	37 countries represented; $\approx$ 91% UK

6. Top 3 Patterns & Business Significance

6.1. Recency is the Dominant Churn Signal

Recency (days since last purchase) shows the strongest correlation with the churn label ($|r| \approx 0.65$). Active customers have a mean recency of **32 days** vs. **214 days** for churned customers (median: 25 vs. 205 days). *Business implication:* A simple recency-based trigger—an automated re-engagement email at day 60—could preemptively reach at-risk customers before they pass the churn threshold.

6.2. Frequency × Monetary Identifies VIP Customers

Active customers average **121 invoices** and **£2,726** in spend vs. **33 invoices** and **£714** for churned customers. Frequency and monetary spend are highly correlated ($r \approx 0.70$). *Business implication:* A loyalty programme targeting the top frequency quartile protects the highest-value segment and maximises retention ROI.

6.3. Geographic Variation in Churn Rate

The dataset covers **37 countries**, with $\approx$ 91% of customers based in the UK. Several non-UK markets exhibit churn rates 10–20 percentage points above the overall mean. *Business implication:* Country-specific retention strategies (localised communications, free-return promotions for high-churn markets) are warranted before broad international marketing investment.